Year 8 SEAL Science · Energy · Revision Worksheet

Name: _____ Class: _____

Instructions

This worksheet has 8 topics and 24 questions (80 marks). Each topic has three questions: **Easy**, then **Medium**, then **Hard**. Work through the topics in order, and do the questions in each topic in order, from Easy to Hard.

Show your working for every calculation and put a unit on every final answer. When you describe a pattern, quote data.

Try every question before you look at the worked solutions. Then mark your own work, and record your score for each topic in the table at the end of the solutions. Revise the topics where you lost the most marks first.

A calculator is allowed.

Formulas

Quantity	Formula
Kinetic energy	KE = ½mv^2 (m in kg, v in m/s, KE in J)
Gravitational potential energy	GPE = mgh (g = 10 N/kg, h in m, GPE in J)
Efficiency	efficiency = useful energy out ÷ total energy in × 100%
Electrical energy	energy (Wh) = power (W) × time (h); 1 kWh = 1,000 Wh
Running cost	cost = energy (kWh) × tariff ($ per kWh)

Topic 1 – Forms of energy, kinetic and gravitational potential energy

Question 1 · Easy (2 marks)

The school bus. The nine types of energy are: kinetic, thermal, light, sound, electrical, gravitational, elastic, chemical and nuclear. Each type belongs to one of two groups: kinetic or potential.

a) For each example, name the type of energy and state whether it is a kinetic or a potential type.

i. The diesel fuel in the school bus's fuel tank.

Type: _____ Kinetic or potential: _____

ii. The noise of the bus horn.

Type: _____ Kinetic or potential: _____

2 marks

Question 2 · Medium (3 marks)

Sports day: long jump. A long jumper has a mass of 50 kg. She takes off at a speed of 7 m/s.

a) Calculate her kinetic energy at take-off. Show your working.

2 marks

b) A jumper with twice her mass takes off at the same speed. How does that jumper's kinetic energy compare with hers? Use the formula to explain.

1 mark

Question 3 · Hard (4 marks)

Sports day: high jump. A high jumper has a mass of 45 kg. During her jump, her body rises 1.2 m higher than it was at take-off. Use g = 10 N/kg.

a) Calculate the gravitational potential energy she gains.

1 mark

b) Assume that all of the kinetic energy of her upward movement at take-off becomes this gravitational potential energy. Calculate her upward take-off speed. Give your answer to one decimal place.

2 marks

c) Her real upward take-off speed must be bigger than your answer to b). Explain why.

1 mark

Topic 2 – Heat transfer

Question 4 · Easy (2 marks)

Around the school.

a) Each sentence describes heat moving. Write *conduction*, *convection* or *radiation* for each.

i. Energy from the Sun travels through space and warms the metal playground slide.

Method: _____

ii. Inside, cool air from the classroom air conditioner sinks and spreads across the room.

Method: _____

2 marks

Question 5 · Medium (3 marks)

The school bus seat. The school bus is parked outside on a cold night. The metal handle and the plastic seat of each bus seat have been in the cold bus all night, so they are at the same temperature. In the morning, a student touches both. The metal handle feels much colder.

a) Explain why the metal handle feels colder than the plastic seat, even though both are at the same temperature.

3 marks

Question 6 · Hard (4 marks)

The old canteen fridge. The canteen's old drinks fridge has no fan. Its freezer compartment is at the **top** of the fridge. Even so, the bottom shelf of the fridge gets cold.

a) Explain how the whole fridge is cooled. Use the words *particles* and *density* in your answer.

3 marks

b) A staff member packs drinks tightly onto every shelf, right up to the freezer, with no gaps. The bottom shelf now stays warmer. Suggest why.

1 mark

Topic 3 – Transfers, transformations, efficiency and Sankey diagrams

Question 7 · Easy (2 marks)

Around the school. A *transfer* moves energy from one object to another. A *transformation* changes energy from one type to another.

a) Write *transfer* or *transformation* for each.

i. The school's solar panels change sunlight into electricity. _____

ii. A student's warm hands warm up the cold metal chain of a swing. _____

2 marks

Question 8 · Medium (3 marks)

The school bus engine. The school bus's engine is 30% efficient. To speed up from one bus stop, the bus needs 540 kJ of useful kinetic energy.

a) Calculate the chemical energy the diesel must supply, and the energy that is wasted.

3 marks

Question 9 · Hard (6 marks)

The school's solar panels. The school compares an old solar panel on the hall roof with a new, larger one. A student also copied the figures for a third panel, Panel X, from an advertisement. The table shows the energy for each panel in one second of full sun.

Panel	Sunlight energy in (J)	Electrical energy out (J)
Old panel	1,600	240
New panel	2,500	550
Panel X	1,800	2,160

a) Calculate the efficiency of the old panel and the new panel. State which of these two is more efficient.

2 marks

b) Sketch a Sankey diagram for one second of the **new** panel. Label each arrow with its energy type and its amount in joules. Each arrow's width should match the energy it shows. Assume all the wasted energy becomes thermal energy.

2 marks

c) The figures for Panel X cannot be right. Use a calculation to explain why.

2 marks

Topic 4 – First Nations fire knowledge and energy changes

Question 10 · Easy (2 marks)

The yarning circle. In the school garden's yarning circle, a visiting First Nations ranger describes ways of starting a fire without matches.

a) Name the fire-starting method in each description.

i. A hard stick is pushed quickly back and forth along a groove in soft wood.

Method: _____

ii. A stone such as flint is struck against ironstone to make hot sparks.

Method: _____

2 marks

Question 11 · Medium (3 marks)

Ember to flame. The ranger explains that a glowing ember is not yet a fire. It is placed in a tinder bundle of fine, dry grass and bark, and then blown on gently.

a) Explain, using energy ideas, why the ember is placed in fine, dry tinder and then blown on.

3 marks

Question 12 · Hard (5 marks)

The fire saw. In the fire saw method, one piece of split wood is held still. The edge of another piece of wood is sawn quickly back and forth across it. Hot wood dust collects in a crack and forms an ember, which is then used to light tinder.

a) Complete the energy flowchart for the fire saw. Write the energy type for each box. Label each arrow *transfer* or *transformation*.

Box 1 (the person's muscles): _____ energy

Arrow 1: _____

Box 2 (the wood moving back and forth): _____ energy

Arrow 2 (friction between the two pieces of wood): _____

Box 3 (the hot wood dust): _____ energy

Arrow 3 (the hot ember is placed in the tinder): _____

Box 4 (the burning tinder): _____ energy → _____ and _____

3 marks

b) Jess says: "All of the energy in the campfire came from the person's arms." Evaluate Jess's claim.

2 marks

Topic 5 – Appliances, star ratings and running costs

Question 13 · Easy (2 marks)

The canteen fridge. The motor (compressor) of the canteen fridge uses 140 W. It runs for a total of 15 hours each day.

a) Calculate the energy the compressor uses each day, in kWh.

2 marks

Question 14 · Medium (3 marks)

The canteen urn. The canteen's hot-water urn uses an average of 2,200 W. It is switched on for 5 hours on each school day, and there are 200 school days in a year. Electricity costs 30 cents per kWh.

a) Calculate the energy the urn uses in a year, in kWh, and its yearly running cost.

3 marks

Question 15 · Hard (5 marks)

A new pie warmer. The canteen's old pie warmer uses an average of 1,500 W. A new pie warmer has insulated walls and double-glazed doors, and uses an average of 900 W to keep the same pies hot. Either warmer would run for 6 hours a day for 200 school days a year. Electricity costs 30 cents per kWh. The new warmer costs $950.

a) Calculate how much money the new warmer would save each year.

2 marks

b) Calculate how many years it would take for the savings to pay back the $950.

1 mark

c) The canteen only buys an appliance if it pays for itself within 4 years. What is the highest price the canteen could pay for the new warmer? Does the $950 warmer meet this rule?

2 marks

Topic 6 – Designing an energy-efficient building

Question 16 · Easy (2 marks)

The school hall and change rooms.

a) The hall needs new ceiling insulation. The supplier offers batts rated R2.0, R3.5 and R5.0. Which is the best insulator? Give a reason.

1 mark

b) The sports change rooms have a solar hot-water system on the roof. The storage tank sits above the collector. Hot water rises from the collector up into the tank by itself, with no pump. Name this method of heat transfer.

1 mark

Question 17 · Medium (3 marks)

The school garden. The windows of the Year 8 classroom block face west. Students plant a row of deciduous trees (trees that lose their leaves in autumn) on the west side of the block.

a) Explain how these trees help keep the classrooms cool in summer but still let them warm up in winter. Name the method of heat transfer involved.

3 marks

Question 18 · Hard (5 marks)

Hot water for the canteen. The canteen's electric storage water heater uses 21 kWh of electricity on each school day. There are 200 school days in a year. A heat-pump water heater would give the canteen the same hot water, but would use only one third of the electricity. Electricity costs 30 cents per kWh.

a) Calculate the yearly cost of running the electric storage water heater.

2 marks

b) Calculate the yearly cost of the heat-pump water heater, and how much money it would save each year.

2 marks

c) Explain why a heat-pump water heater needs less electricity than an electric storage water heater to heat the same water.

1 mark

Topic 7 – Investigating: variables, patterns, errors and improvements

Questions 19 to 21 all use the same investigation.

Question 19 · Easy (2 marks)

Wax on rods. Lena tests how well different materials conduct heat. She uses four rods of the same length and thickness: copper, steel, glass and wood. She sticks a small drop of wax onto each rod, 6 cm from one end. She stands that end of the rod in a beaker of boiling water and times how long it takes the wax drop to melt and fall off. She does three trials for each rod and stops each trial at 600 s.

Lena's hypothesis: *If a rod is made of metal, then its wax drop will fall off sooner, because metals are good conductors of heat.*

Rod material	Trial 1 (s)	Trial 2 (s)	Trial 3 (s)	Average (s)
Copper	48	52	50	50
Steel	182	245	178	
Glass	did not fall	did not fall	did not fall	more than 600
Wood	did not fall	did not fall	did not fall	more than 600

You will calculate the steel average in Question 20.

a) Identify the independent variable and the dependent variable in Lena's investigation.

Independent variable: _____

Dependent variable: _____

2 marks

Question 20 · Medium (3 marks)

Use Lena's results table above.

a) Lena thinks one of her repeat trials went wrong. Which result is the outlier? Give a reason, using the other trials.

2 marks

b) Calculate the average time for the steel rod. Leave out the outlier.

1 mark

Question 21 · Hard (5 marks)

Use Lena's results table and hypothesis from Question 19.

a) Describe the pattern in the average times. Does the data support Lena's hypothesis? Quote data in your answer.

2 marks

b) Explain, using particles, why the wax on the copper rod fell off first.

1 mark

c) Suggest one change to Lena's method that would improve her investigation. Explain how your change would improve her results.

2 marks

Topic 8 – Energy audits and balancing supply and demand

Question 22 · Easy (3 marks)

The school energy audit. Questions 22 to 24 all use this audit. Year 8 students audit the school's electricity use on a typical school day. The school uses 750 kWh of electricity a day.

Use	Heating and cooling	Lighting	Computers and IT	Canteen and kitchen	Hot water	Other
Share (%)	36	24	20	8	7	5

a) Identify the use that takes the largest share of the school's electricity.

1 mark

b) Write *behaviour*, *appliance* or *building* beside each change.

i. Closing classroom doors when the heating is on. _____

ii. Fitting double glazing to the hall windows. _____

iii. Replacing the old canteen urn with a better-insulated urn. _____

2 marks

Question 23 · Medium (3 marks)

On a sunny day, the school's solar panels supply 480 kWh. Demand is 750 kWh.

a) Calculate how much electricity the school must buy from the grid that day.

1 mark

b) Calculate the percentage of the school's demand that the solar panels supply.

1 mark

c) Explain why the school must stay connected to the grid, even on sunny days.

1 mark

Question 24 · Hard (5 marks)

The school council is looking at two proposals. Use the audit table and the school's demand of 750 kWh a day.

Proposal 1: set the thermostats 2 °C lower in winter and 2 °C higher in summer. This would cut heating and cooling energy by 20%. **Proposal 2:** fit motion sensors to classroom lights. This would cut lighting energy by 30%.

a) Calculate the energy each proposal would save each day, in kWh.

2 marks

b) The council can only choose one proposal this year. Evaluate which proposal it should choose. Justify your answer.

2 marks

c) If the school did both proposals, would the 480 kWh from the solar panels on a sunny day now meet its demand? Show a calculation.

1 mark

END OF WORKSHEET